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# Observability Debt in Agent Evaluation: Toward a Telemetry Layer for AgenticOS

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## Abstract

1     Agentic AI systems are evaluated today through three disjoint public infrastructures—  
2     leaderboards, trace corpora, and telemetry logs—none of the public sources we exam-  
3     ine jointly exposes the fields a resource-aware operating layer would need: task identity,  
4     model identity, scaffold and policy-configuration identity, outcome, token/cache usage, re-  
5     tries, latency, and billing. We measure its consequences across three studies on public  
6     data. First, using the Holistic Agent Leaderboard (HAL), raw dollar-cost rankings trans-  
7     fer across some benchmark pairs (Spearman  $\rho=0.74\text{--}0.93$ ) but are mostly absorbed by a  
8     workload-persistent displayed-label cost propensity (pooled model  $R^2=82.0\%$ ); success-  
9     adjusted cost is markedly less portable (mean  $\rho=0.59$  vs.  $0.84$ ). Second, using 977 exact  
10    task–model–matched trajectory pairs from Open-SWE-Traces, scaffold choice changes ob-  
11    servable execution burden at fixed task and model (OpenHands/SWE-agent turn ratio  $0.85$ ,  
12     $[0.82, 0.87]$ ), and the widely assumed “expensive failure” premium is not detected under  
13    exact task matching (80%-power threshold: premium fractions  $\geq 0.65$ ; observed  $0.44\text{--}$   
14     $0.49$ ). Third, auditing two independent telemetry corpora (8,058 and 1,017 sessions), we  
15    characterize the resource anatomy—93.7% mean cache-read share,  $4.29\times$  conditional tool-  
16    error cascades replicated across sources, top-1%-of-sessions holding 48.3% of input-token  
17    volume—that a scheduler or router would need but that no benchmark currently attaches  
18    to an outcome. These studies are complementary rather than linked: Studies 1–2 show  
19    that transferred raw-cost rankings and unconditioned failure premiums confound *which*  
20    *workloads were attempted* with *how agents behaved on them*, while Study 3 shows that  
21    the execution variables hidden beneath those aggregates are themselves heterogeneous,  
22    heavy-tailed, and state-dependent. From this evidence we derive a joint telemetry schema  
23    spanning identity, execution, resources, control, and outcome, and argue it belongs at the  
24    operating-system layer: controlling execution and allocating resources requires observing  
25    what those allocations depend on.

## 26   1   Why This Belongs at the OS Layer

27    Agentic AI systems persist state, invoke tools, retry failed actions, and accumulate context across long horizons.  
28    Reasoning about their resource behavior—for scheduling, routing, cost budgeting, or governance—requires the  
29    same unified telemetry that operating systems have long provided for conventional workloads: a common  
30    schema linking *what ran*, *on what*, *under which policy*, *with what outcome*, and *at what resource cost*. We  
31    use *observability debt* to name the accumulated inability of public agent-evaluation infrastructure to jointly  
32    attribute resource consumption to task, model, execution policy, outcome, and system-level events. In every  
33    public source we examined, this debt prevents reliable attribution of observed resource differences to workload,  
34    model, scaffold, configuration, or execution behavior.

35    What makes this an *operating-system* responsibility rather than better benchmark bookkeeping? An agentic  
36    OS occupies a specific functional position. It observes *identity* (which task, model, scaffold, and policy con-  
37    figuration is running), *execution* (which tools act, what context accumulates), *resources* (tokens, cache, time,  
38    memory, money), *control* (retry, terminate, route, budget), and *outcome* (did the task succeed). The same  
39    resource-accounting principle motivates an OS-level treatment: the layer that controls execution and allocates  
40    resources needs access to the telemetry required to make those decisions. Without per-round token components,  
41    a memory manager cannot budget context growth; without configuration identity, a router cannot learn *why* one

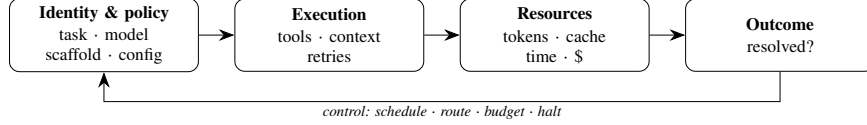


Figure 1: **The agentic OS control loop.** Each stage consumes exactly the schema fields derived in Section 6; the feedback arrow hosts scheduling, routing, budgeting, and halting decisions that today run on aggregate dollars because no examined public source closes this loop (Table 2).

label is cheap; without error-status streams, a governor cannot halt a spiraling session. Today, each of these decisions is made implicitly from aggregate dollar figures (Figure 1).

We treat three public data sources as stand-ins for what this observability layer would unify: a cost-and-outcome leaderboard (HAL), a task/model/scaffold-matched trajectory corpus (Open-SWE-Traces), and raw session-level usage telemetry (two independent corpora). Each exposes a different partial view. HAL links dollar cost to outcome but encrypts execution traces. Open-SWE-Traces exposes exact task/model/scaffold matching and resolved status but no tokens or billing. The telemetry corpora expose token, cache, and tool-call dynamics but no outcome or billing. **No public source we examined jointly exposes task, model, scaffold and policy configuration, outcome, token/cache usage, retries, latency, and billing**, the joint schema a resource-aware OS layer needs (Table 2). Importantly, combining these sources does not create record-level observability: their different execution units, identifiers, and release boundaries prevent resource use, outcome, and billing from being joined into a single per-execution record.

Our three studies are therefore complementary rather than linked—each probes a different dimension of the gap, and none can be joined to another at the record level. (1) A coverage-aware cost-portability study (HAL) shows that aggregate dollar cost conflates displayed-label propensity with benchmark difficulty and decorrelates under success adjustment. (2) A matched-trajectory study shows that execution structure differs measurably across scaffolds, while a popular failure-cost heuristic turns out to measure workload composition. (3) A cross-source telemetry audit shows that the execution variables hidden beneath aggregate cost are themselves highly heterogeneous, heavy-tailed, and state-dependent. Together they triangulate one gap; no pair of them jointly identifies a causal mechanism. This paper is an empirical requirements study for an AgenticOS telemetry layer, rather than an implementation of the layer itself.

## 2 Related Work

Kapoor et al. (2026) introduce HAL as cost-aware evaluation infrastructure spanning coding, web, science, and customer-service tasks (21,730 rollouts, roughly \$40,000 in evaluation cost). Kapoor et al. (2024) motivate joint accuracy–cost reporting and convex-hull frontier analysis, which we adopt as one of three decision-frontier definitions. Ndzomga (2026) asks whether task subsets preserve agent rankings within a single benchmark; we instead ask whether an *observed cost signal* predicts a *different* workload. Benchmark<sup>2</sup> studies cross-benchmark rank consistency for static LLM benchmarks; our setting adds interactive scaffolds, dollar cost, success adjustment, and a held-out workload test. SWE-Effi reports that failed attempts consume substantially more resources than successful ones; our matched design tests that claim at fixed task and does not detect the premium once task-difficulty confounding is removed (Section 4). Open-SWE-Traces supplies the exact matching that makes this test possible but lacks token/billing fields, motivating Section 6. Adjacent work sits on either side of this seam: commercial LLM tracing platforms expose per-deployment span usage but are not cross-benchmark evaluation infrastructure, while public evaluation releases strip execution detail; serving-side schedulers and KV-cache managers presuppose exactly the per-request visibility our audited sources lack. We quantify this seam on public data rather than propose another tracer or scheduler.

## 3 Study 1: Does a Cost Signal Travel Across Workloads?

### 3.1 Data and cohort

We use the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga (2026): 242 displayed evaluations across nine benchmarks, 11 scaffolds, and 29 displayed model labels. The CSV supplies accuracy and aggregate dollar cost but no token components, provider aliases, tool-call counts, retries, or per-task trajectories—the exact gap this paper is about.

The only scaffold repeated broadly enough is the displayed HAL Generalist Agent. Fixing it and matching exact label strings yields six workloads with adequate overlap—CORE-Bench Hard, GAIA, SciCode, ScienceAgent-Bench, SWE-bench Verified Mini, TAU-bench Airline—for a cohort of 86 rows and 25 labels (USACO has one Generalist observation and is excluded).

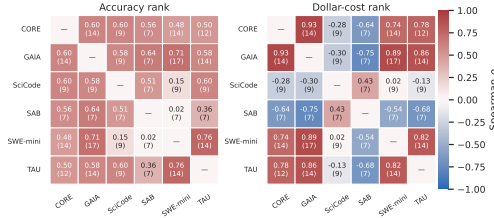


Figure 2: **Raw cost transfers; success-adjusted cost does not.** Pairwise Spearman rank correlations across benchmark pairs sharing  $\geq 12$  displayed labels. Raw dollar cost is strongly portable (mean 0.84); cost per expected success is not (mean 0.59), and only three CPS pairs survive Holm-corrected permutation testing.

### 3.2 Raw cost transfers; the mechanism behind it does not

Across six pairs with  $\geq 12$  shared labels, raw dollar-cost Spearman  $\rho$  ranges 0.74–0.93 (mean 0.84), surviving Holm-corrected permutation testing (Figure 2). A pooled label-plus-benchmark log-cost fixed-effects model explains  $R^2=82.0\%$  (adjusted 72.7%); once label and benchmark propensity are removed, **no high-overlap residual-cost association survives** a 20,000-draw permutation test.

Leave-one-benchmark-out (LOBO) prediction succeeds for CORE ( $\rho=0.78$ ,  $N=14$ ,  $p=0.0016$ ), GAIA (0.87,  $N=16$ ), SWE-mini (0.90,  $N=16$ ), TAU (0.83,  $N=14$ ), all robust to every leave-one-label-out omission (e.g., CORE [0.73, 0.91], SWE-mini [0.88, 0.92]). It fails for SciCode (unstable,  $[-0.50, 0.14]$ ; 2.1% mean accuracy, one-third zero-accuracy labels) and ScienceAgentBench (directionally negative,  $[-0.77, -0.43]$ ; narrowest log-cost dispersion,  $N=7$ ).

**Interpretation:** the portable component behaves as a workload-persistent displayed-label propensity—plausibly reflecting provider pricing, reasoning-mode defaults, or routing policy—that a router could in principle learn and reuse. but public data cannot identify *which*, since the distinguishing fields are not exposed.

### 3.3 Success-adjusted cost is a different, less portable ranking

Cost per expected success,  $CPS_{i,b}(\epsilon) = C_{i,b} / \max(A_{i,b}, \epsilon)$  with primary floor  $\epsilon=0.01$  (sensitivities at 0.001, 0.05, and zero-exclusion in the supplement), has mean  $\rho=0.59$ . Only three pairs survive correction: CORE–GAIA (0.87, [0.63, 0.96]), CORE–TAU (0.83, [0.31, 1.00]), GAIA–TAU (0.78, [0.42, 0.94]). The clearest raw-vs-CPS gap is GAIA–SWE-mini (0.60, [0.14, 1.13]): a label cheap in dollars is not reliably cheap per expected success elsewhere. No label tested on  $\geq 5$  workloads sits on every accuracy–cost frontier under discrete nondominance, convex hull, or 5% tolerance (o4-mini Low reaches 5/6 under tolerance; no universal winner under any strict definition).

**Implication:** routing on displayed dollar cost optimizes a signal that decorrelates from efficiency exactly where procurement decisions matter.

## 4 Study 2: What Matched Trajectories Reveal That Aggregates Cannot

HAL’s own trace archives are encrypted, so Study 1 cannot say *why* costs differ. We use an external unencrypted corpus—977 exact task–model pairs from OpenSWE-Traces for MiniMax-M2.5 under two scaffolds (OpenHands, SWE-agent), a reproducible but nonrandom four-shard boundary sample. The schema exposes resolved status, roles, tool calls, and text but not billed tokens or dollars; we treat turns, tool calls, and characters as trace-burden proxies.

### 4.1 Scaffold choice changes observable burden at fixed task

OpenHands has lower matched medians: 123 vs. 149 trajectory turns (ratio 0.85, [0.82, 0.87]), 61 vs. 74 tool calls (0.85, [0.82, 0.86]), 176,708 vs. 194,263 characters (0.95, [0.92, 0.98]). Resolved status shows no detected difference ( $p=0.197$ ): scaffolds differ in execution structure without a detected accuracy trade-off here.

### 4.2 The “expensive failure” premium is not detected under exact task matching

Comparing failed vs. resolved trajectories *across* tasks reproduces SWE-Eff’s pattern (19–36% heavier failures). But among the **107 discordant pairs**—identical task, identical model, one scaffold resolved and the other failed—the failed side is heavier only 47–52 times out of 107 on every metric (two-sided binomial  $p \geq 0.25$ ; ob-

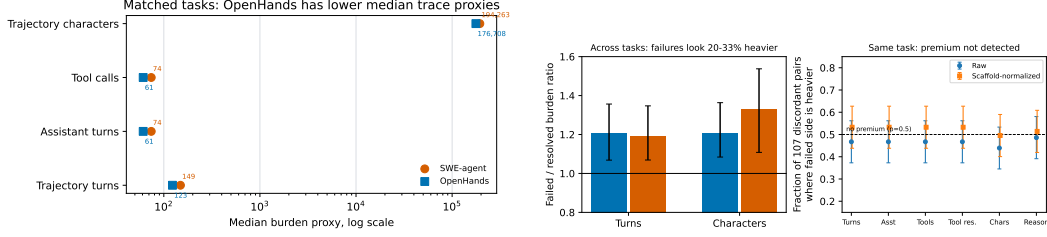


Figure 3: **(a)** Scaffold burden on 977 exact task–model pairs: OpenHands uses  $\sim 15\%$  fewer median turns/tool calls than SWE-agent at fixed task and model. **(b)** The failure premium is not detected under exact task matching (80% power against premiums  $\geq 15\text{pp}$ ). Left: naive cross-task ratios reproduce folklore (19–36% heavier failures). Right: among 107 discordant pairs, the failed side is heavier only 47–52/107 on every metric.

Table 1: **Telemetry anatomy across two independent corpora.** Cache metrics are computed within each corpus; a share and a total-volume ratio are different quantities and not interconvertible. Conditional-error ratios are computed on unrounded per-call rates.

|                                                                    | TraceLab                                  | MIMO Claude Code                     |
|--------------------------------------------------------------------|-------------------------------------------|--------------------------------------|
| Sessions / LLM rounds                                              | 8,058 / 665,453                           | 1,017 / 4,690                        |
| Median input / output tokens per round                             | 132,092 / 249                             | n/a (constant-tuple audit)           |
| Cache-read vs. uncached input                                      | 93.7% mean share ( $\approx 14.9\times$ ) | 753.0M / 46.3M tokens = $16.3\times$ |
| Median final-vs-initial context                                    | $3.6\times$ (P90 $9.9\times$ )            | n/a                                  |
| Top-1% / top-10% input-token share                                 | 48.3% / 88.2%                             | — / 37.9%                            |
| Error cascade $P(\text{err} \text{err})/P(\text{err} \text{succ})$ | $21.0/4.9\% = 4.29\times$                 | $4.3\%$ vs. $1.0\%$ ( $4.33\times$ ) |
| Errored calls retried same-tool / retry succeeds                   | 88.3% / 77.9%                             | —                                    |
| Consecutive-error run length (median/P90/max)                      | 1 / 3 / 368                               | —                                    |
| Tool-call latency, success vs. error (median)                      | 140 ms / 313 ms                           | —                                    |
| Tool-result size, success vs. error (median chars)                 | 602 / 208                                 | —                                    |

served fraction CIs  $[0.37, 0.57]$  raw,  $[0.43, 0.63]$  normalized). A pre-specified power calculation gives 80% power against a premium fraction  $\geq 0.65$  (failed heavier on 70+ of 107); observed 0.44–0.49 rules out strong versions but not modest ones ( $< 15\text{pp}$ ). One model, nonrandom sample: this tests the heuristic here, not its general absence.

**Interpretation:** hard tasks make agents both burn more *and* fail more. An unconditioned failure multiplier measures *which tasks were attempted*, not a property of failing runs. Task-conditioned accounting requires exactly the joint schema no single public source provides.

## 5 Study 3: The Telemetry an OS Layer Would Need to See Directly

Study 1 shows that aggregate cost cannot explain *why* an agent is expensive; Study 2 shows that execution structure varies across agent systems at fixed task and model. This section shows that the execution variables hidden beneath aggregate cost are themselves highly heterogeneous, heavy-tailed, and state-dependent. Setting aside outcome and billing entirely: what does raw execution telemetry look like? We audit two independent coding-agent corpora (Table 1). **TraceLab:** 8,058 real-world sessions, 665,453 LLM rounds, 743,819 tool calls. **MIMO Claude Code:** 1,017 sessions, 4,690 rounds, single model. Neither links usage to outcome or billing; all summaries are compositional.

**Context is the dominant resource, not generation.** The median TraceLab round carries 132,092 input tokens against 249 output tokens (93.7% mean cache-read share); median final-round context is  $3.6\times$  initial (P90  $9.9\times$ ). A leaderboard dollar total cannot distinguish price-driven from cache-policy- or accumulation-driven spend. Tool verbosity drives this: median next-input growth is 443 / 1,412 / 5,420 tokens below 1k / 1k–10k / above 10k tool-result characters.

**Resource burden is heavy-tailed.** The top 1% of sessions hold 48.3% of summed input volume; top 10% hold 88.2%. Median session input rises from 0.34M tokens at zero observed tool errors to 14.65M at six or more; mean-based budgeting would miss this tail-driven concentration (Figure 4).

**Error cascades replicate across independent sources.** After a successful tool call, the next call errs 4.9%; after an error, 21.0% ( $4.29\times$ ). MIMO—different model, different site—replicates at  $4.33\times$  ( $4.3\%$  vs.  $1.0\%$ ). Same-tool retry is common (88.3% of errored calls get one; 77.9% succeed), but the tail is extreme: consecutive-error runs have median length 1, P90 3, max 368. Failures are also slower yet less informative:

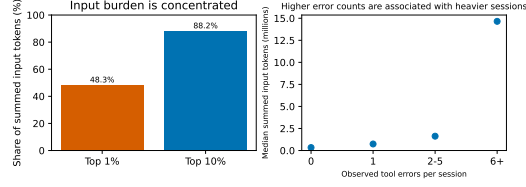


Figure 4: **Heavy-tailed burden.** Top-1%/top-10% concentration of summed input tokens (left axis) and median session input by observed-error bin (right): 0.34M at zero errors rising to 14.65M at 6+.

Table 2: **The observability gap.** Field coverage per public source we examined: ■ exposed, □ partial (display-label-only, degraded fidelity, or inferable), – absent. No examined source covers the joint schema; the union still leaves latency unlinked and round-level token fidelity unverifiable outside one corpus.

| Field                           | HAL | Open-SWE | TraceLab | MIMO |
|---------------------------------|-----|----------|----------|------|
| Task identity                   | □   | ■        | –        | –    |
| Model identity                  | □   | ■        | □        | ■    |
| Scaffold identity/version       | ■   | ■        | □        | ■    |
| Configuration / policy identity | □   | –        | –        | –    |
| Outcome / resolved status       | ■   | ■        | –        | –    |
| Tokens / cache components       | –   | –        | ■        | □    |
| Retries / error status          | –   | □        | ■        | □    |
| Latency                         | –   | –        | ■        | –    |
| Billing                         | ■   | –        | –        | –    |

153 median latency 313 ms vs. 140 ms (P95 85.6 s vs. 15.8 s) while returning 208 vs. 602 characters—more wall  
154 time, less signal.

155 **Data-quality boundary.** MIMO repeats one constant per-round usage tuple, so its within-session token dy-  
156 namics are not validated; we use it only for session counts, totals, concentration, and content-derived error  
157 sequences.

## 158 6 An Evidence-Derived Joint Telemetry Schema

159 Table 2 shows the coverage gap; the missing rows are precisely those Sections 3–5 show to be load-bearing.  
160 Each field is tied to an analysis above or to the OS control function it unblocks; *evidence-derived* means we  
161 claim usefulness and necessity for those functions, not formal minimality. Keyed to the control function each  
162 feeds:

- 163 1. (*Attribution / exact matching*) **Task and model identity** (exact IDs, not display strings)—enables exact-  
164 matched designs; without it the Section 4 test is infeasible on public data.
- 165 2. (*Routing between scaffolds*) **Scaffold/harness identity and version**—attributes burden differences to scaf-  
166 fold design rather than model behavior; input to routing.
- 167 3. (*Configuration attribution*) **Configuration and policy identity** (reasoning mode, sampling params, bud-  
168 gets, routing/cache policy)—distinct from model identity; Study 1’s unexplained propensity component is  
169 plausibly the shadow of these missing fields.
- 170 4. (*Success-conditioned accounting*) **Outcome/resolved status**—conditions any cost statistic on success, clos-  
171 ing the CPS portability gap.
- 172 5. (*Memory / budget management*) **Token components per round (input, output, cache-read, cache-  
173 creation)**—separates price from accumulation and cache policy; lets a memory layer budget context growth.
- 174 6. (*Governance / halting*) **Tool-call counts, retries, error status per call**—detects and prices  $4.29\times$  cascade  
175 dynamics; a design motivation for governance policies that halt spiraling sessions—motivated here, not  
176 experimentally validated.
- 177 7. (*Real-time scheduling*) **Latency per call and per session**—present only as unlinked tool-call durations in  
178 one corpus; required for real-time scheduling.
- 179 8. (*Procurement / auditing*) **Billing, linked to all of the above**—the one field HAL exposes well, and exactly  
180 the field that is uninterpretable without the others.

181 Figure 1 places these fields at the stages of the control loop they close; Table 2 shows why, on public data alone,  
182 the loop does not close.

183 No public source we examined exposes this joint schema. We do not propose a storage format or API—a  
184 systems-design question outside our scope—but the measured consequences of its absence argue it belongs at

185 a shared operating-system layer rather than being re-derived piecemeal inside each benchmark, where today it  
186 is not derived at all.

## 187 7 Limits

188 This is a secondary-data analysis with clear construct boundaries. Display labels do not fully identify prompts,  
189 tools, budgets, or harness versions; the HAL cohort is sparse, unbalanced, mostly single-run. Propensity adjust-  
190 ment is observational, not list-price control; CPS is an aggregate ratio, not the mean cost of successful runs.  
191 Study 2 uses one model and a nonrandom four-shard sample whose proxies are not tokens or dollars, powered  
192 only against strong premiums ( $\geq 15$ pp). Study 3 is observational and unlinked to outcomes or billing; MIMO's  
193 per-round fields are unverifiable and used only where supported, and its design implications (tail-aware bud-  
194 getting, cascade-triggered halting) are motivations, not evaluated policies. None of this invalidates the schema  
195 argument—each limit is itself an instance of the missing-field problem.

## 196 8 Conclusion

197 Across three complementary studies, we find that commonly used aggregate resource signals cannot be reliably  
198 attributed to workload, execution, or outcome without telemetry fields that the public evaluation sources we  
199 examined do not jointly expose. Raw-cost rankings transfer across workloads but largely reflect a workload-  
200 persistent displayed-label cost propensity; success-adjusted cost reorders the ranking (surviving correction for  
201 three of six pairs); the widely assumed failure-cost premium is not detected under exact task matching. Beneath  
202 all three results sits one missing artifact: no examined public source jointly links task, model, scaffold and con-  
203 figuration, outcome, token/cache usage, retries, latency, and billing. Because the layer that controls execution  
204 and allocates resources is the natural owner of exactly this telemetry, we argue the schema belongs at a shared  
205 operating-system layer for agentic AI—the role resource accounting plays in conventional OSES—rather than  
206 being reconstructed independently, incompletely, and incompatibly inside each leaderboard or trace release.

## 207 References

- 208 W. U. Ahmad, N. Ludwig, S. Majumdar, and B. Ginsburg. Open-  
209 SWE-traces: Advancing dual-mode multilingual distillation for  
210 software engineering agents. *arXiv preprint arXiv:2606.16038*,  
211 2026.  
212 Z. Fan, K. Vasilevski, D. Lin, et al. SWE-effi: Re-evaluating software  
213 AI agent system effectiveness under resource constraints. *arXiv  
214 preprint arXiv:2509.09853*, 2025.
- 215 S. Kapoor et al. AI agents that matter. *arXiv preprint  
216 arXiv:2407.01502*, 2024.  
217 S. Kapoor et al. Holistic agent leaderboard: The missing infrastruc-  
218 ture for AI agent evaluation. *arXiv preprint arXiv:2510.11977*,  
219 2026.  
220 Franck Ndzomga. Efficient benchmarking of AI agents. *arXiv  
221 preprint arXiv:2603.23749*, 2026.  
222 Q. Qian, C. Huang, et al. Benchmark<sup>2</sup>: Systematic evaluation of  
223 LLM benchmarks. *arXiv preprint arXiv:2601.03986*, 2026.

## 224 NeurIPS Paper Checklist

- 225 1. **Claims** [Yes]Empirical, scoped to public data; confounds, proxy limits, power stated (Sec. 3–5, Limits). [Yes]LLM usage  
226 disclosed (13).  
227 2. **Limitations** [Yes]Single-model matched sample, nonrandom traces, proxies not dollars, observational telemetry (Sec. 7,  
228 Sec. 5 boundary). [Yes]CIs/binomial intervals throughout.  
229 3. **Theory** [N/A]Observational fixed-effects decompositions; assumptions stated (Sec. 3), no proofs.  
230 4. **Reproducibility** [Yes]Pipeline + frozen outputs released; inputs public. [N/A]No training.  
231 [Yes]Permutation/Holm/bootstrap + power (Sec. 3–4).  
232 5. **Open Access** [Yes]On publication; inputs already public. [Yes]Maintained repo.  
233 6. **Setting** [N/A]No training; CPS floor  $\epsilon$  and permutation counts stated. [Yes]Public CSV/trace corpora, processed locally.  
234 7. **Statistical Significance** [Yes]Intervals as above. [Yes]Holm-corrected permutation; binomial sign tests with pre-specified  
235 power.  
236 8. **Societal Impact** [Yes]Improved eval infrastructure; no direct harm; gaps are missing-field problems, no harmful applica-  
237 tion enabled.  
238 9. **Safeguards** [N/A]No dangerous artifacts; public evaluation traces.  
239 10. **Licenses** [Yes]All inputs under public research licenses (HAL, Open-SWE-Traces, TraceLab, MIMO).  
240 11. **New Assets** [Yes]Analysis pipeline, figures, proposed schema (Sec. 5–6).  
241 12. **Crowdsourcing / Human Subjects** [N/A]None; no IRB needed.  
242 13. **LLM Usage** [Yes]LLM assistants aided drafting/editing; all claims and numbers verified by authors against code and data.